Week 2: Stationarity, ARIMA Specification and Diagnostics for ONS UK Vacancies

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Abstract

This report applies the methodology from my Week 1 review to the real ONS vacancy series (VACS02, all industries, 3-month rolling average, thousands). Using only a chronological training sample (June 2001 – September 2024, 280 observations), with the last 24 observations held out untouched, I (1) test for stationarity and decide on the order of differencing, (2) compare 50 ARIMA$(p, 1, q)$ models (plus a handful of robustness variants) using ACF/PACF plots, AIC, AICc and BIC, and (3) examine the residuals of the recommended model. **Recommendation:** ARIMA$(1, 1, 2)$ **without drift, fitted to the series in levels.** Differencing once is clearly needed; the model is the best on all three information criteria; the Ljung–Box test does not reject at 6 or 12 lags, but there is modest, isolated residual autocorrelation at lags 10, 13, 16 and 24 (all below 0.16 in absolute value) and the residual variance is strongly inflated by the 2020–21 COVID-19 episode. I explain below what this means for the forecasting stage.

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1 Data and set-up

1.1 The series

The workbook `vacs02oct2026.xlsx` has two sheets: "levels" (vacancies, thousands) and "job openings rate". I use the *levels* sheet, column `AP2Y` ("All vacancies", SIC 2007 sections B–S), because the Week 1 review and this brief refer to the vacancy count itself. Three facts about this series matter for everything that follows.

1. **It is a 3-month rolling average, published monthly.** A label such as "Jul–Sep 2026" is the average of July, August and September. I index each observation by the *last* month of its window (so "Jul–Sep 2026" → September 2026). Consecutive observations therefore share two of their three underlying months. This is the point flagged under "Points to discuss (b)" in Week 1, and it has a concrete statistical consequence (Section 3.1): the differenced series has built-in autocorrelation of moving-average type.
2. **It is not seasonally adjusted, but the ONS states that it shows no seasonality** (footnote 2 of the sheet), so I do not use seasonal terms. I check this indirectly through the ACF in Section 3.1 and the residual ACF in Section 4: there is no spike at lags 12 or 24 that would indicate a seasonal cycle.
3. **The data run from April–June 2001 to July–September 2026**, giving 304 consecutive monthly observations (June 2001 to September 2026) with no gaps. One label in the spreadsheet contains a typo ("Nov- Jan 2002", with a stray space) which a naive parser would drop, silently losing a month; my script's pattern is tolerant of this and a check asserts that the month sequence has no gaps.

1.2 Chronological split

As requested, model selection uses only the earlier observations.

	Window (month ending)	Observations
Training (all tests, selection, diagnostics)	June 2001 – September 2024	280
Held out (not touched in this report)	October 2024 – September 2026	24

The 24-month hold-out is long enough to evaluate 1-, 3- and 6-month-ahead forecasts at several origins in Week 3. One caveat: because of the rolling average, the first two hold-out values

(windows ending October and November 2024) still contain August–September 2024 data that are inside the training sample. This is a property of the data, not of my split, and I flag it so that it is considered when the forecasting evaluation is designed. No test, plot-based decision or information criterion below uses any hold-out value; the hold-out is shown in grey in Figure 1 only for orientation.

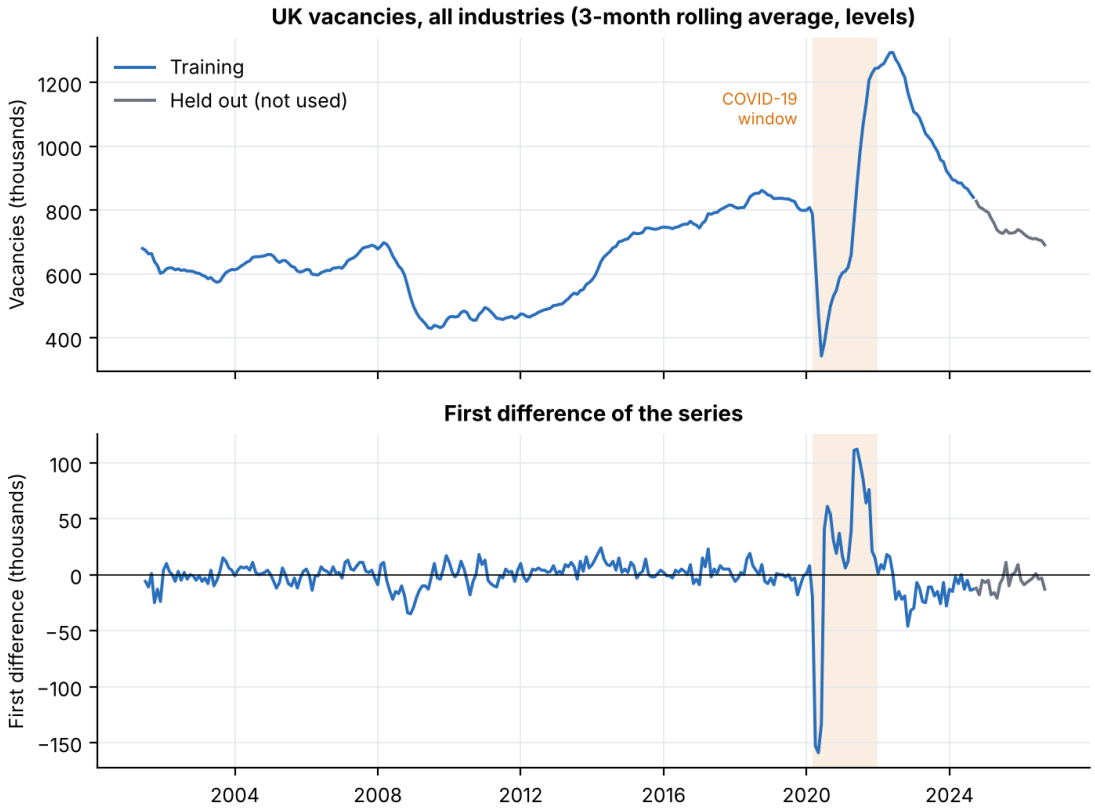


Figure 1: Top: the vacancy series (levels, thousands), with the training sample in blue and the held-out period in grey. The shaded band marks March 2020 – December 2021. Bottom: the first difference. The series wanders with no fixed mean (a 2009 collapse, a long recovery, the 2020 plunge, a record 1.29 million in mid-2022, then a decline), while its first difference fluctuates around zero except in the COVID window.

1.3 Software and reproducibility

Everything is produced by one script, `analysis.py` (Python 3.13, `statsmodels` 0.15, `scipy`, `pandas`, `matplotlib`), run as `python3 analysis.py <vacs02.xlsx> <output dir>`. It writes every figure, every table (as CSV and as LaTeX) and the numerical results quoted below. ARIMA models are fitted by exact Gaussian maximum likelihood (`statsmodels.tsa.arima.model.ARIMA`); for $d = 1$ the first observation initialises the differenced process with a diffuse prior, so the first residual is a start-up artefact and is dropped from all residual diagnostics, and the likelihood of all models is computed on the same 279 differenced observations, which makes the information criteria comparable.

2 Task 1: Stationarity and differencing

2.1 What is being tested and why two tests

A series is (weakly) stationary if its mean, variance and autocovariances do not depend on time. ARIMA's AR and MA terms estimate fixed patterns, so they need a stationary series to work with; differencing ($d > 0$) is the remedy for a unit root.

Augmented Dickey–Fuller (ADF). Regress Δy_t on a constant (and optionally a linear trend), on y_{t-1} and on k lagged differences:

$$\Delta y_t = \alpha + \beta t + \gamma\, y_{t-1} + \sum_{i=1}^{k} \delta_i \Delta y_{t-i} + e_t.$$

The null hypothesis is a *unit root* ($\gamma = 0$); a large negative t-statistic on γ rejects it, i.e. supports stationarity. The lagged differences absorb autocorrelation in e_t. I report two choices of k (chosen by AIC and by BIC), since the answer can be sensitive to it, and two deterministic specifications (constant only; constant and trend).

KPSS. KPSS reverses the roles: its null is that the series is *stationary* (around a level, or around a trend), and a large statistic rejects it in favour of a unit root. The two tests have opposite nulls, so using both gives a confirmatory check. The useful reading is:

ADF	KPSS	Conclusion
rejects unit root	does not reject stationarity	stationary
does not reject	rejects	unit root, difference the series
both reject / neither rejects		inconclusive (e.g. outliers, low power)

As the Week 1 review put it, the outcome is a guide, not a verdict, so I also look at plots and at the ACF.

2.2 Results on the original and differenced series

Table 1 gives the tests on the training sample.

Table 1: Unit-root and stationarity tests on the training sample (June 2001 – September 2024).

	ADF (H_0: unit root)				KPSS (H_0: stationary)	
Series	const, AIC	const, BIC	c+trend, AIC	c+trend, BIC	level	trend
Level	−1.37 (0.595)	−2.89 (0.046)	−2.38 (0.388)	−3.85 (0.014)	1.26 (≤0.01)	0.31 (≤0.01)
1st difference	−5.67 (<0.001)	−4.00 (0.001)	−5.67 (<0.001)	−3.98 (0.009)	0.06 (≥0.10)	0.04 (≥0.10)
2nd difference	−5.43 (<0.001)	−11.11 (<0.001)	−5.43 (<0.001)	−11.10 (<0.001)	0.11 (≥0.10)	0.10 (≥0.10)
log Level	−2.46 (0.126)	−2.26 (0.184)	−3.46 (0.044)	−2.93 (0.152)	1.18 (≤0.01)	0.31 (≤0.01)
log 1st difference	−4.65 (<0.001)	−6.21 (<0.001)	−4.64 (<0.001)	−6.19 (<0.001)	0.07 (≥0.10)	0.04 (≥0.10)

Each cell: test statistic (p-value). "AIC"/"BIC" = rule used to choose the number of augmenting lags. ADF 5% critical values are about −2.87 (constant) and −3.42 (constant + trend). KPSS p-values are interpolated and truncated to [0.01, 0.10].

Original (level) series.

- The ADF test with AIC-chosen lags (15 lags) does *not* reject a unit root (statistic −1.37, $p = 0.595$; with a trend −2.38, $p = 0.388$).

- With BIC-chosen lags (7 lags) the ADF test rejects narrowly ($p = 0.046$ with a constant, $p = 0.014$ with a trend).
- KPSS rejects stationarity decisively under both nulls (statistics 1.26 and 0.31 against 5% critical values 0.463 and 0.146; $p \leq 0.01$).
- The lag-1 autocorrelation of the level is 0.993, and the ACF decays only very slowly (Figure 2, top left). An AR(1) with constant fitted to the levels gives $\hat{\phi} = 0.991$ (s.e. 0.007), indistinguishable from the unit-root boundary.

The two ADF variants disagree, so the ADF alone would be borderline. The evidence nevertheless points one way. KPSS rejects strongly, the slowly decaying ACF is the textbook symptom of a unit root, and the ADF rejection under BIC lags is fragile: it appears or disappears with the lag rule. A plausible reason is the 2020–21 V-shaped collapse and rebound, which looks like fast mean reversion to the ADF regression and can inflate its power to reject. The pre-COVID sub-sample test below supports this reading.

First difference. Every ADF variant rejects a unit root ($p \leq 0.009$; statistics between -3.98 and -5.67), and KPSS does not reject stationarity (statistics 0.057 and 0.043, both below the 10% critical values). Both tests agree, so the first-differenced series is stationary in the sense that matters for ARIMA modelling.

Second difference. Also stationary on every test, as expected: a stationary series stays stationary when differenced again. That does not mean $d = 2$ is needed. Needing a second difference would require evidence that the *first difference* still has a unit root, and the tests above reject exactly that.

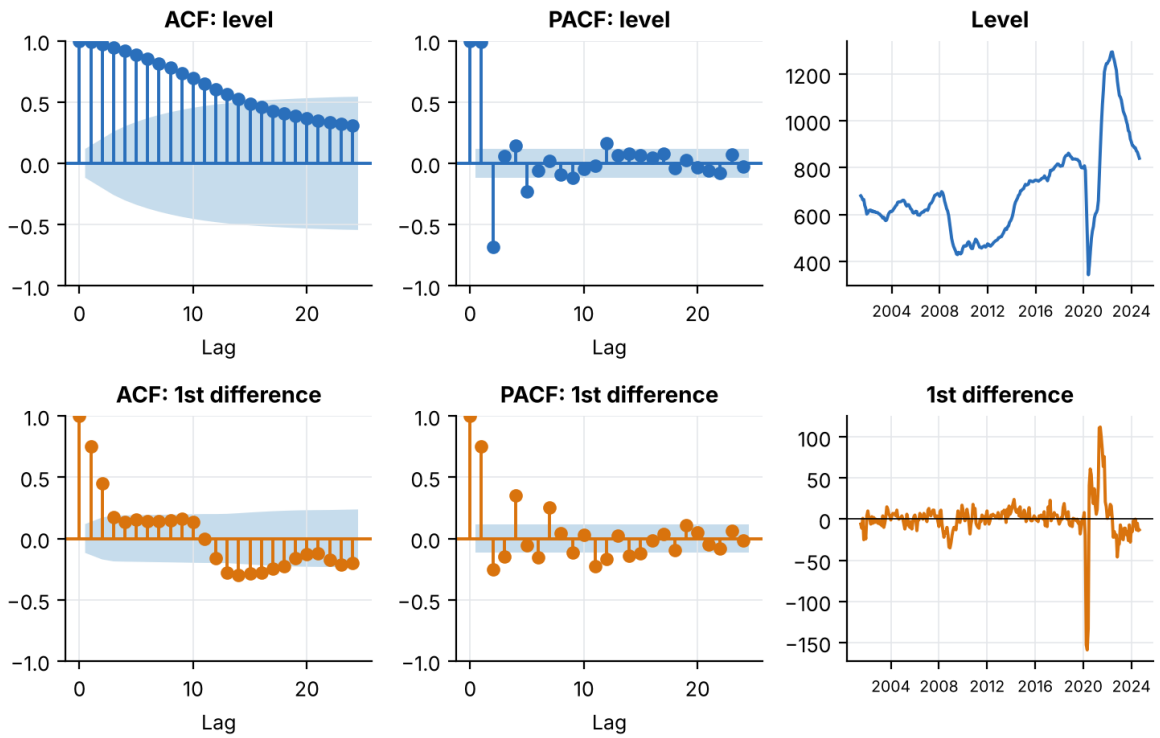


Figure 2: ACF and PACF of the training series in levels (top) and first differences (bottom). Shaded bands are approximate 95% intervals. The level ACF decays slowly (non-stationary); the differenced ACF drops quickly, with the PACF showing a large first partial autocorrelation.

2.3 Is first differencing enough, or too much?

A common check for over-differencing is whether the differenced series has a large negative lag-1 autocorrelation (near -0.5) or a larger variance than before. Here:

	Level ($d = 0$)	First difference	Second difference
Standard deviation	198.8	24.2	17.1
Lag-1 autocorrelation	0.993	0.750	+0.098

The second difference has a *positive* lag-1 autocorrelation, so there is no over-differencing signature. The standard deviation does fall slightly from $d = 1$ to $d = 2$, which the "choose the d with minimum variance" rule of thumb would read as favouring $d = 2$. I do not follow that rule here, because the first difference is itself strongly autocorrelated (lag-1 correlation 0.75, a consequence of the 3-month averaging), so differencing it again mechanically removes some variance without that being a sign of a unit root. The formal tests (ADF rejects, KPSS does not reject, for Δy_t) are the better guide and they say $d = 1$.

The first differences also look stable over time apart from the COVID window (Figure 1, bottom, and Figure 3).

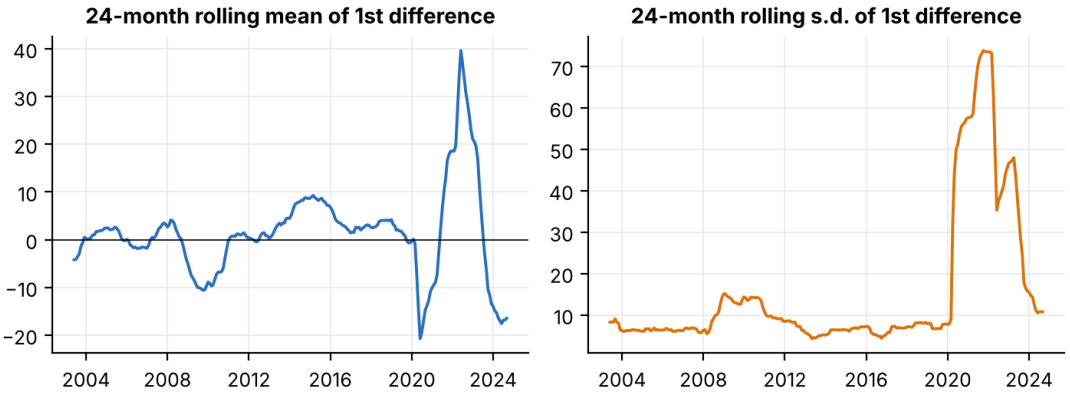


Figure 3: 24-month rolling mean and standard deviation of the first difference. The mean stays near zero apart from the COVID swing; the spread is stable except for that episode.

2.4 Robustness: log scale and sub-samples

Table 2: Sub-sample tests. The post-COVID sample has only 33 observations, so these tests have little power.

	ADF (H_0: unit root)				KPSS (H_0: stationary)	
Series	const, AIC	const, BIC	c+trend, AIC	c+trend, BIC	level	trend
Pre-COVID level (to Dec 2019)	−1.40 (0.583)	−1.37 (0.598)	−1.89 (0.660)	−1.73 (0.738)	0.82 (≤0.01)	0.45 (≤0.01)
Pre-COVID 1st diff	−3.50 (0.008)	−4.19 (<0.001)	−3.55 (0.035)	−4.14 (0.006)	0.30 (≥0.10)	0.09 (≥0.10)
Post-COVID level (from Jan 2022)	−1.05 (0.734)	0.46 (0.984)	−1.99 (0.606)	−2.40 (0.382)	0.74 (0.010)	0.11 (≥0.10)
Post-COVID 1st diff	−3.28 (0.016)	−3.03 (0.032)	−2.27 (0.452)	−3.13 (0.100)	0.17 (≥0.10)	0.15 (0.043)

Each cell: test statistic (p-value). "AIC"/"BIC" = rule used to choose the number of augmenting lags. ADF 5% critical values are about −2.87 (constant) and −3.42 (constant + trend). KPSS p-values are interpolated and truncated to [0.01, 0.10].

- **Log scale** (Table 1, bottom rows): the conclusion is the same, and slightly cleaner. The ADF fails to reject a unit root in the log level under three of four specifications ($p =$

0.126, 0.184, 0.152; the exception is $p = 0.044$ with trend and AIC lags), KPSS rejects, and the log first difference is stationary on every test.

- **Pre-COVID sample** (June 2001 – December 2019, 223 observations): free of the 2020–21 shock, the level series is clearly non-stationary (ADF $p = 0.58$ and 0.60 with a constant, $p = 0.66$ and 0.74 with a trend; KPSS rejects), and its first difference is stationary (ADF $p \leq 0.035$; KPSS does not reject). This supports the interpretation above: the unit-root conclusion is not an artefact of COVID; it is the borderline ADF rejection on the full sample that is.
- **Post-COVID sample** (from January 2022, 33 observations): too short for the tests to have power, and results are mixed (e.g. KPSS with trend rejects for the differenced series, $p = 0.043$). I report it for completeness but give it no weight.

Conclusion for Task 1. Use $d = 1$. The level series is non-stationary (KPSS rejects, slowly decaying ACF, AR(1) coefficient of 0.991, pre-COVID ADF cannot reject), the first difference is stationary on all ADF and KPSS variants, and a second difference is unnecessary (no over-differencing signature; stationarity of Δy_t is not rejected). The conclusion is the same on the log scale and without the COVID period.

3 Task 2: Comparing ARIMA specifications

3.1 Reading the ACF and PACF of the differenced series

With $d = 1$, Figure 2 (bottom) gives the starting points for p and q (band ± 0.117 for 279 observations):

Lag	1	2	3	4	5	6
ACF of Δy_t	0.750	0.451	0.173	0.136	0.153	0.142
PACF of Δy_t	0.750	−0.255	−0.146	0.354	−0.053	−0.155

The ACF falls off sharply after lag 2 (from 0.45 to 0.17), which suggests an MA component of order about 2. The PACF has several significant spikes (lags 1, 2, 3, 4, ...), so a pure AR model would need many terms. These are only suggestions, but there is a structural reason to take $q = 2$ seriously: if the underlying *monthly* series had uncorrelated changes ε_t, its 3-month moving average would have $\Delta y_t = (\varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2})/3$, which is exactly an MA(2) process (theoretical autocorrelations 2/3 and 1/3 at lags 1 and 2, zero afterwards). The observed values (0.75, 0.45) are larger, consistent with that MA(2) structure *plus* genuine persistence in monthly changes, which an AR term can capture. This points to a small ARMA$(p, 2)$ model for Δy_t.

3.2 Candidate set and criteria

I fitted ARIMA$(p, 1, q)$ for every $0 \leq p, q \leq 4$, each with and without a drift term (a constant in the differenced series, i.e. a linear trend in the level): 50 models, all on the same 280 observations and same d, so that their information criteria are comparable. The criteria are

$$\text{AIC} = -2\ell + 2k, \qquad \text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}, \qquad \text{BIC} = -2\ell + k \ln n,$$

where ℓ is the maximised log-likelihood, k the number of estimated parameters (including the innovation variance) and $n = 279$ the number of differenced observations. Smaller is better. BIC

penalises extra parameters more heavily than AIC and is therefore the more conservative guide, which was the reasoning in Week 1 for short samples; AICc corrects AIC for small n/k. As a rule of thumb, a difference of 0–2 between two models means no meaningful preference, 4–7 is clear evidence, and more than 10 is strong. Information criteria measure fit net of complexity; they do not say whether a model's residuals are clean, so I use the Ljung–Box test as a second criterion (the adequacy check requested in Task 3), with its degrees of freedom reduced by $p + q$.

I do not compare models with different d by information criteria (not valid, because the differenced data differ), which is why d was settled first by the tests in Task 1.

3.3 Comparison table

Table 3 compares a short list: the naive random walk $(0, 1, 0)$ as the floor, the standard low-order models, and the leading models from the full grid. The complete 50-model grid is in `tables/arima_grid_full.csv` and the top 15 by BIC are in Appendix A.

Table 3: Candidate ARIMA$(p, 1, q)$ specifications on the training sample ($n = 280$, 279 differenced observations). Δ = difference from the best value in the full 50-model grid; bold Ljung–Box p-values are below 0.05.

Model	k	AIC	AICc	BIC	ΔAIC	ΔBIC	$\hat{\sigma}$	LB(6)	LB(12)	LB(24)
ARIMA(0,1,0)	1	2570.0	2570.0	2573.6	322.6	311.7	24.12	**0.000**	**0.000**	**0.000**
ARIMA(0,1,1)	2	2419.1	2419.1	2426.4	171.7	164.5	18.33	**0.000**	**0.000**	**0.000**
ARIMA(1,1,1)	3	2332.2	2332.3	2343.1	84.9	81.2	15.62	**0.000**	**0.000**	**0.000**
ARIMA(0,1,2)	3	2254.8	2254.9	2265.7	7.4	3.8	13.55	**0.005**	**0.008**	**0.000**
ARIMA(1,1,2)*	4	2247.4	2247.5	2261.9	0.0	0.0	13.32	0.474	0.199	**0.006**
ARIMA(2,1,2)	5	2249.2	2249.5	2267.4	1.9	5.5	13.32	0.282	0.151	**0.005**
ARIMA(0,1,3)	4	2248.0	2248.2	2262.6	0.7	0.7	13.33	0.367	0.151	**0.003**
ARIMA(1,1,3)	5	2249.0	2249.2	2267.2	1.7	5.3	13.31	0.223	0.135	**0.007**
ARIMA(1,1,4)	6	2249.6	2249.9	2271.4	2.3	9.5	13.28	0.264	0.148	**0.009**
ARIMA(1,1,2)+drift	5	2249.3	2249.6	2267.5	2.0	5.6	13.32	0.473	0.199	**0.006**

k counts all estimated parameters including the innovation variance. LB(m): Ljung–Box p-value using m lags and $m - p - q$ degrees of freedom. * recommended. ARIMA(0,1,0) has no estimated mean, so $k = 1$ (the variance) and LB uses no degree-of-freedom correction.

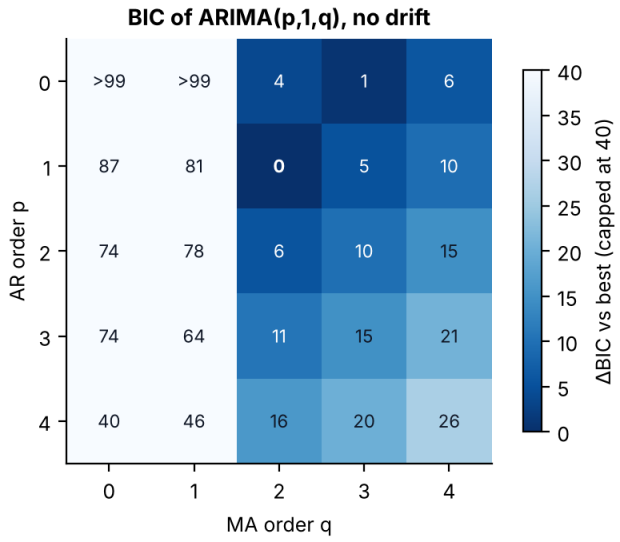


Figure 4: ΔBIC relative to the best model for each (p, q) with $d = 1$ and no drift. Models with $q \leq 1$ are far worse (the MA(2) structure is essential); beyond $q = 2$ and $p = 1$, extra terms do not help.

What the evidence says.

- $q \geq 2$ **is essential.** Going from $(0, 1, 1)$ to $(0, 1, 2)$ lowers BIC by about 161 points; $q \leq 1$ models are rejected by every criterion (Figure 4) and have Ljung–Box p-values of essentially zero. This matches the ACF reading and the moving-average structure of the data.
- ARIMA$(1, 1, 2)$ **is best on all three criteria** (AIC 2247.4, AICc 2247.5, BIC 2261.9). Adding an AR(1) term to the $(0, 1, 2)$ model improves BIC by 3.8 and AIC by 7.4 and fixes the short-lag residual autocorrelation (LB(6) rises from 0.005 to 0.47, LB(12) from 0.008 to 0.20).
- ARIMA$(0, 1, 3)$ **is a close competitor** (BIC 0.7 higher, AIC 0.7 higher). A ΔBIC of 0.7 is no real distinction, and the model has the same number of parameters. I prefer $(1, 1, 2)$ because it is ranked first by all criteria and because the AR(1) term has a clear interpretation (persistence of monthly changes) on top of the averaging-induced MA(2).
- **More complex models do not earn their keep.** $(2, 1, 2)$, $(1, 1, 3)$ and $(1, 1, 4)$ have BIC 5.3–9.5 higher (clear evidence against) and AIC 1.7–2.3 higher as well, so the extra terms do not improve fit enough to pay for themselves.
- **Drift is not needed.** Adding a drift to $(1, 1, 2)$ increases AIC by 2.0 and BIC by 5.6; the estimated drift is 0.42 thousand per month with $p = 0.91$. This is expected for a series that rises and falls over time.
- **The naive** $(0, 1, 0)$ **model** is more than 300 BIC points worse than the best model, confirming that there is substantial short-run dynamics to exploit.

3.4 Is the choice robust?

Table 4 repeats the $0 \leq p, q \leq 4$ search under alternative treatments of the data.

Table 4: Does the choice of specification survive alternative treatments? Each row repeats the 25-model ARIMA$(p, 1, q)$ grid ($0 \le p, q \le 4$, no drift).

Treatment	Best BIC	Best AIC	ΔBIC of (1,1,2)	LB(12)	LB(24)
Training sample, levels (baseline)	(1,1,2)	(1,1,2)	0.0	0.199	**0.006**
Training sample, log levels	(0,1,2)	(0,1,2)	5.4	0.220	0.348
Pre-COVID only (Jun 2001–Dec 2019)	(1,1,4)	(1,1,4)	6.7	0.432	0.182
Training sample + 4 outlier dummies	(0,1,3)	(0,1,3)	1.6	0.280	**0.008**

LB columns refer to the best-BIC model of that row. Dummies are single-month additive outliers at Apr 2020, Jul 2020, May 2021, Aug 2021 (months with $|z| > 3$ in the baseline residuals; chosen after inspecting residuals, so this is a sensitivity check, not an independent test).

- **Log scale.** The best model is $(0, 1, 2)$, with $(1, 1, 2)$ 5.4 BIC points behind. Raw information criteria from log and level fits are *not* comparable, because the likelihoods refer to different variables. Table 5 converts the log-scale criteria to the scale of the original data (adding $2\sum_t \ln y_t$, the Jacobian of the transformation). On that fair comparison, the level model is better by 168 BIC points: modelling the log series fits the data markedly worse, even though its Ljung–Box test at lag 24 is cleaner (0.35). The log residuals also show very strong volatility clustering around 2020 (ARCH-LM $p < 0.001$), because the log transformation magnifies the proportional size of the 2020 collapse. I therefore stay on the level scale.
- **Pre-COVID only.** The best model becomes $(1, 1, 4)$ and passes the lag-12 and lag-24 Ljung–Box tests; $(1, 1, 2)$ is 6.7 BIC points behind. The qualitative picture, $q \ge 2$ plus a single AR term, is the same, but with the COVID shock out of the sample a longer MA structure is favoured. This is useful information about the stability of the specification: it is possible that the COVID months push the full-sample fit towards the more parsimonious $(1, 1, 2)$.
- **Outlier dummies.** Adding single-month additive-outlier dummies for the four months with $|z| > 3$ (Section 4) makes $(0, 1, 3)$ the best model, with $(1, 1, 2)$ only 1.6 BIC points behind, a statistical tie. The $(1, 1, 2)$ coefficients barely move ($\hat{\phi}_1 = 0.31$ vs. 0.22, $\hat{\theta}_1 = 0.90$ vs. 0.83, $\hat{\theta}_2 = 0.77$ vs. 0.81). Because these months were identified from the baseline residuals, this is a sensitivity check rather than an independent test.

Table 5: Levels versus log levels. Log-scale AIC/BIC have been converted to the scale of the original data by adding $2\sum_t \log y_t$, so the four rows are directly comparable.

Scale	Model	Adj. AIC	Adj. BIC	ΔBIC	LB(12)	LB(24)	ARCH-LM
Levels	ARIMA(1,1,2)	2247.4	2261.9	0.0	0.199	**0.006**	0.135
Levels	ARIMA(0,1,2)	2254.8	2265.7	3.8	**0.008**	**0.000**	0.333
Log levels	ARIMA(0,1,2)	2419.3	2430.2	168.3	0.220	0.348	**0.000**
Log levels	ARIMA(1,1,2)	2421.1	2435.6	173.7	0.164	0.335	**0.000**

3.5 Recommended specification and its estimates

Recommendation: ARIMA$(1, 1, 2)$**, no drift, on the series in levels.**
$(1 - \phi_1 B)(1 - B)\, y_t = (1 + \theta_1 B + \theta_2 B^2)\, \varepsilon_t, \qquad \varepsilon_t \sim (0, \sigma^2).$

Table 6: Maximum-likelihood estimates, ARIMA(1,1,2), no drift, training sample.

Parameter	Estimate	Std. error	z	p-value	95% CI
ϕ_1 (AR 1)	0.218	0.076	2.86	0.004	[0.068, 0.367]
θ_1 (MA 1)	0.831	0.045	18.38	<0.001	[0.742, 0.919]
θ_2 (MA 2)	0.812	0.021	37.96	<0.001	[0.770, 0.854]
σ^2	177.386	5.323	33.33	<0.001	[166.954, 187.818]

All three coefficients are significant. The AR coefficient $\hat{\phi}_1 = 0.22$ is modest: a month's change carries over about a fifth into the next. The MA coefficients (0.83 and 0.81) are close to $(1, 1)$, which is the pattern a 3-month moving average of an underlying series with uncorrelated changes would produce $(1 + B + B^2)$. Both roots of the MA polynomial have modulus 1.11: the model is invertible, but close to the boundary. This is a sign that the "averaging" structure is being picked up almost exactly, and it means that parameter estimates and standard errors should be treated with some caution (near-boundary MA models have non-standard sampling behaviour).

Figure 5 shows the one-step fitted monthly change against the actual change in the training sample since 2012.

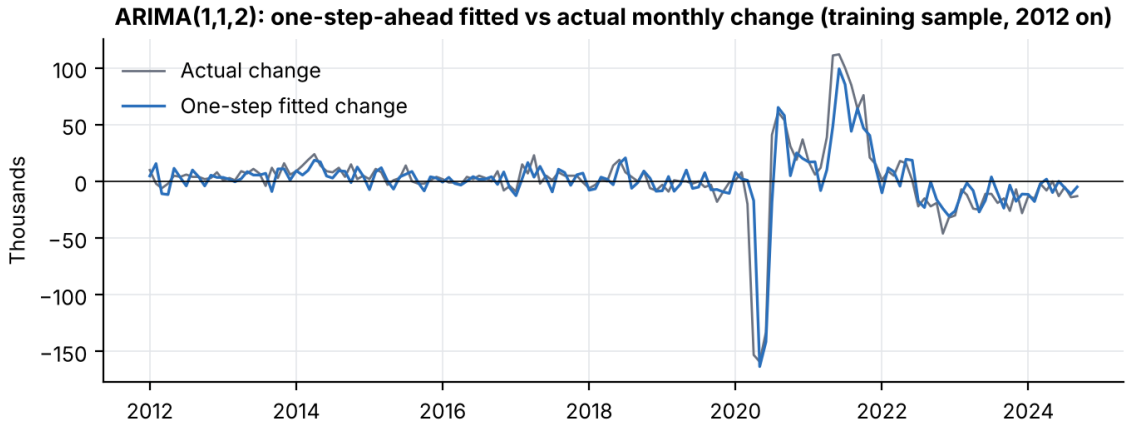


Figure 5: One-step-ahead in-sample fitted vs. actual monthly change in the vacancy series (training sample, 2012 onwards, which includes the COVID shock).

4 Task 3: Model diagnostics

If the model is adequate, its (one-step) residuals should behave like white noise: no remaining autocorrelation, roughly constant variance, and no outliers. Tests and plots look at different aspects of that, so I use both. The residuals are the 279 one-step-ahead prediction errors from the fitted model, with the diffuse first value dropped.

4.1 Residual plots

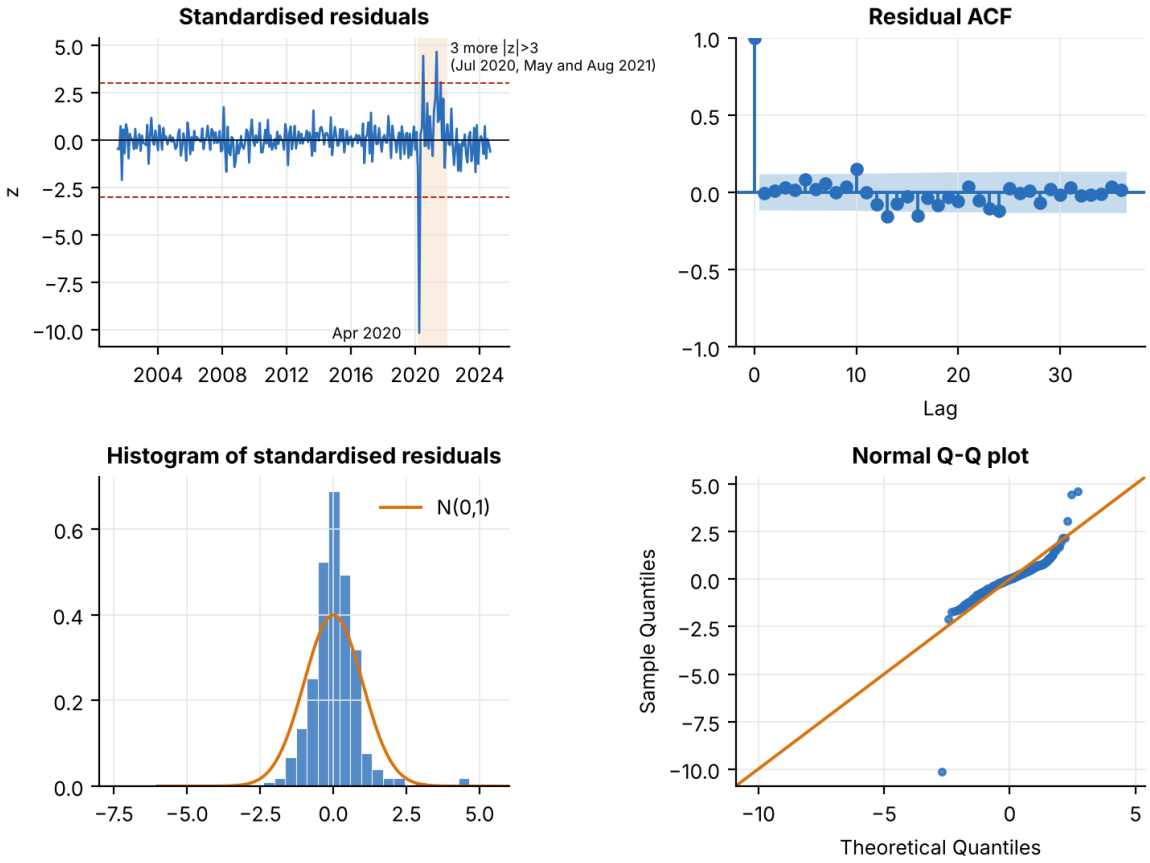


Figure 6: Diagnostics for ARIMA$(1, 1, 2)$: standardised residuals through time (dashed lines at ± 3; the shaded band is March 2020 – December 2021); residual ACF with 95% bands; histogram compared with the standard normal; normal Q–Q plot.

- **Through time:** residuals look like noise around zero before 2020 and after 2022. In the COVID window there are large spikes: April 2020 is a -10.2 standard-deviation event, and July 2020, May 2021 and August 2021 exceed $+3$ (4.4, 4.6, 3.1). These are genuine shocks that no stationary ARIMA could have anticipated.
- **Residual ACF:** almost all bars lie within the band. The first 9 lags are clean (the largest in absolute value is 0.08). Four bars are outside the 95% band: lag 10 ($+0.15$), lag 13 (-0.16), lag 16 (-0.15) and lag 24 (-0.12). With 36 lags examined, about two exceedances would be expected by chance at the 5% level, so four slightly-above-chance exceedances of small size (every $|\hat{\rho}| \leq 0.16$) are a mild finding, not an alarming one. Lag 12 is not significant and lag 24 only just exceeds the band (-0.118 vs. 0.117), so there is no clear seasonal pattern, consistent with the ONS statement.
- **Histogram and Q–Q:** the bulk of the distribution is close to normal but with a very heavy left tail from April 2020. This is a single-observation feature, as the next subsection shows.

4.2 Formal tests

Table 7: Residual diagnostics for ARIMA(1,1,2) (279 residuals; the first, a diffuse start-up value, is dropped).

Test	Statistic	p-value
Ljung–Box, 6 lags (df $= 6-3$)	2.51	0.474
Ljung–Box, 12 lags (df $= 12-3$)	12.26	0.199
Ljung–Box, 18 lags (df $= 18-3$)	30.39	**0.011**
Ljung–Box, 24 lags (df $= 24-3$)	40.51	**0.006**
Ljung–Box, 36 lags (df $= 36-3$)	43.32	0.108
Ljung–Box on squared residuals, 12 lags	10.95	0.534
Engle ARCH-LM, 6 lags	9.76	0.135
Jarque–Bera, all residuals	18814.0	**0.000**
Jarque–Bera, excluding Mar 2020–Dec 2021	6.5	**0.039**

Ljung–Box. The statistic $Q(m) = n(n+2)\sum_{j=1}^{m} \hat{\rho}_j^2/(n-j)$ tests the null that the first m residual autocorrelations are jointly zero; under the null it is χ^2 with $m-p-q$ degrees of freedom. Results:

- Lags 6 and 12: no evidence of residual autocorrelation ($p = 0.47$ and 0.20). This is the range that matters for 1-, 3- and 6-month forecasts, and it is also the range in which models with fewer terms clearly fail, e.g. $(0, 1, 2)$ has $p = 0.005$ at lag 6.
- Lags 18 and 24: rejection at 5% ($p = 0.011$ and 0.006). Figure 7 shows the p-value as a function of the number of lags: it is high up to $m \approx 12$, is mostly below 0.05 for m between about 13 and 30 because of the small spikes at lags 10, 13, 16 and 24, and recovers to $p = 0.11$ at $m = 36$.
- Competing models do no better: $(1, 1, 3)$ and $(1, 1, 4)$ show the same pattern and $(0, 1, 2)$ fails at every lag (Figure 7). Fitted to the pre-COVID sample only, $(1, 1, 2)$ fails even at lag 12 ($p < 0.001$; lag 24: $p = 0.006$), so this feature is in the data, not just a COVID artefact. It is not removed by dummies for the four COVID outliers either (LB(24) $p = 0.009$ with dummies).

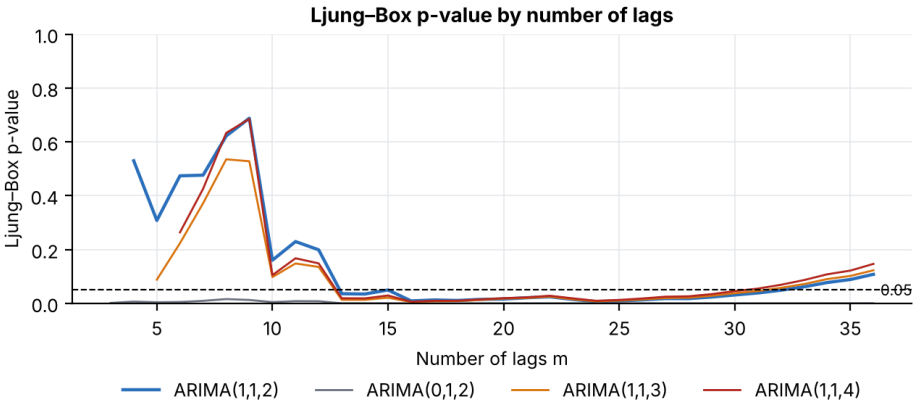


Figure 7: Ljung–Box p-value against the number of lags m for four candidates. All pass at short lags and dip below 0.05 for m from about 13 to 30; $(0, 1, 2)$ fails at every lag.

Heteroskedasticity. The residual standard deviation depends heavily on the period:

Period	Before Mar 2020	Mar 2020 – Dec 2021	From Jan 2022
Observations	224	22	33
Residual s.d. (thousands)	7.8	38.7	10.8

During the COVID window the residual s.d. is nearly five times its pre-2020 value. Even so, the standard ARCH tests do not flag it: Engle's ARCH-LM test with 6 lags gives $p = 0.135$ and the Ljung–Box test on squared residuals (12 lags) $p = 0.53$. I would not read that as clean. ARCH tests look for volatility that clusters and persists, whereas here the volatility is concentrated in one regime of 22 months dominated by a single extreme observation, which is a different (and for these tests less detectable) pattern. The regime table is the more informative evidence. The post-2022 s.d. (10.8) is also higher than the pre-2020 value (7.8), partly because the level of vacancies is now higher and the changes larger.

Normality. Over all residuals, normality is strongly rejected (Jarque–Bera = 18,814; skewness -3.02; excess kurtosis 39.8), but this is driven almost entirely by April 2020. Excluding March 2020 – December 2021, the residuals are close to normal (skewness -0.28, excess kurtosis 0.55; Jarque–Bera 6.5, $p = 0.039$, still a marginal rejection). The Durbin–Watson statistic is 2.00, a further sign of no first-order autocorrelation.

4.3 Is the specification appropriate for our research?

Verdict: adequate as a working specification, with three caveats.
What is good: the mean structure is captured. Ljung–Box does not reject at lags 6 and 12, the models with fewer terms are clearly worse, and the parameter values have a sensible interpretation.
Caveats: (i) small residual autocorrelations at lags 10, 13, 16 and 24 (all $|\hat{\rho}| \leq 0.16$), which make the Ljung–Box test reject at lags 18–24 and which no candidate model removes; (ii) variance that is about five times larger during the COVID window, so the constant-variance assumption fails; (iii) non-normal residuals driven by April 2020.

How much this matters depends on how the model is used. For *point forecasts* at horizons of 1–6 months, the small long-lag correlations are unlikely to matter much, because the model's information is in the first few lags. For *prediction intervals*, however, the estimated innovation s.d. of 13.3 is a blend of a quiet regime (≈ 8) and the COVID regime (≈ 39): Gaussian intervals built on it would be too wide in calm periods and too narrow in a crisis. For the ARIMAX stage, the unmodelled COVID shock could also be wrongly attributed to the Indeed index.

5 Conclusions and suggested next steps

1. **Differencing:** $d = 1$. The level is non-stationary and the first difference is stationary (ADF and KPSS agree); no second difference is warranted.
2. **Specification:** ARIMA$(1, 1, 2)$ without drift, on the level of vacancies. It is first on AIC, AICc and BIC among 50 candidates, and the MA(2) component is a direct and expected consequence of the series being a 3-month rolling average. ARIMA$(0, 1, 3)$ is a statistical near-tie and is the natural alternative to carry into Week 3.
3. **Diagnostics:** Ljung–Box does not reject up to lag 12 (one isolated spike at lag 10); mild autocorrelation at lags 10–24 not removed by any candidate; strongly heteroskedastic residuals during COVID; heavy left tail from April 2020.

Suggested next steps for the forecasting stage (these are suggestions for discussion, not things I have done):

- Report forecast accuracy for ARIMA$(1,1,2)$ and ARIMA$(0,1,3)$ against the naive random walk on the held-out 24 months, re-estimating at each origin, and report the COVID treatment explicitly (full sample versus pre-COVID versus outlier dummies).
- Use empirical or regime-aware prediction intervals, since the residual variance is not constant.
- Allow for the overlap in the rolling average when setting the first forecast origins in the hold-out sample (see Section 1.2).
- Re-run the specification search on the pre-COVID sample alongside the full sample: it favours a longer MA structure $(1,1,4)$ and is a useful check of stability.

Limitations. (a) The search over 50 models and the use of the same data for selection and diagnostics means that reported p-values are conditional on selection and are optimistic. (b) The COVID treatment uses dummies chosen after looking at the residuals. (c) The rolling-average structure means observations are not independent monthly measurements; the model describes the *published smoothed series*, not the underlying monthly vacancy count. (d) Near-non-invertibility of the MA(2) part means parameter standard errors are approximate. (e) I did not model the job-openings-rate sheet or any other industry series.

A Top 15 models by BIC

Table 8: Top 15 of the 50 grid models ranked by BIC.

Model	k	AIC	AICc	BIC	LB(12)	LB(24)	Converged
ARIMA(1,1,2)	4	2247.4	2247.5	2261.9	0.199	**0.006**	yes
ARIMA(0,1,3)	4	2248.0	2248.2	2262.6	0.151	**0.003**	yes
ARIMA(0,1,2)	3	2254.8	2254.9	2265.7	**0.008**	**0.000**	yes
ARIMA(1,1,3)	5	2249.0	2249.2	2267.2	0.135	**0.007**	yes
ARIMA(2,1,2)	5	2249.2	2249.5	2267.4	0.151	**0.005**	yes
ARIMA(1,1,2)+drift	5	2249.3	2249.6	2267.5	0.199	**0.006**	yes
ARIMA(0,1,4)	5	2249.6	2249.8	2267.7	0.129	**0.003**	yes
ARIMA(0,1,3)+drift	5	2250.0	2250.2	2268.2	0.151	**0.003**	yes
ARIMA(0,1,2)+drift	4	2256.8	2256.9	2271.3	**0.008**	**0.000**	yes
ARIMA(1,1,4)	6	2249.6	2249.9	2271.4	0.148	**0.009**	yes
ARIMA(2,1,3)	6	2249.8	2250.1	2271.6	0.142	**0.009**	yes
ARIMA(3,1,2)	6	2250.8	2251.1	2272.5	0.118	**0.005**	yes
ARIMA(1,1,3)+drift	6	2251.0	2251.3	2272.8	0.136	**0.007**	yes
ARIMA(2,1,2)+drift	6	2251.2	2251.5	2273.0	0.151	**0.005**	yes
ARIMA(0,1,4)+drift	6	2251.5	2251.9	2273.3	0.129	**0.003**	yes

B Files delivered

- `analysis.py`: full script (tests, 50-model grid, diagnostics, robustness, figures, tables).

- `tables/arima_grid_full.csv`: all 50 fitted models (AIC, AICc, BIC, ΔIC, Ljung–Box p-values, convergence); `tables/shortlist.csv`: the comparison table of Table 3; `tables/stationarity_tests.csv`: every ADF and KPSS result; `tables/robustness.csv`; `tables/levels_vs_log.csv`; `tables/recommended_residuals.csv`.
- `figures/`: the seven figures used above.